3. Algebraic Fractions

Lowest Common Multiple (L.C.M.)

Lowest common Multiple of simple polynomials can be easily found resolved into their elementary factors

Example 1

Find the L.C.M of $8x^3$, $12x^5$, and $18x^7$

$8x^3 = 2^3 \times x^3$

$12x^5 = 2^2 \times 3 \times x^5$

$18x^7 = 2 \times 3^2 \times x^7$

Hence L.C.M. is $2^3 \times 3^2 \times x^7 = 72x^7$

Example 2

Find the L.C.M of $2x^2 - 8$, $3x^2 + 3x - 6$ **and** $6x^2 - 6x - 12$

$2x^2 - 8 = 2(x^2 - 4) = 2(x-2)(x+2)$

$3x^2 + 3x - 6 = 3(x^2 + x - 2) = 3(x+2)(x-1)$

$6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x-2)(x+1)$

L.C.M is $6(x-2)(x+2)(x-1)(x+1)$

Simplying Algebraic Fractions

Example 1 $\dfrac{2}{x^2-1} - \dfrac{3}{(x-1)^2}$

We have to find L.C.M of $(x^2 - 1)$ and $(x-1)^2$

$x^2 - 1 = (x-1)(x+1)$

$(x-1)^2 = (x-1)^2$

L.C.M. is $(x-1)^2 (x+1)$

$$\frac{2}{x^2-1} - \frac{3}{(x-1)^2}$$

$$= \frac{2}{(x-1)(x+1)} - \frac{3}{(x-1)^2}$$

$$= \frac{2(x-1)-3(x+1)}{(x-1)^2(x+1)}$$

$$= \frac{-x-5}{(x-1)^2(x+1)}$$

$$= \frac{-(x+5)}{(x-1)^2(x+1)}$$

Example 2

Simplify.

$$\frac{2}{1+x}+\frac{1}{x-1}+\frac{3x}{1-x^2}$$

$$= \frac{2}{1+x}-\frac{1}{1-x}+\frac{3x}{1-x^2}$$

$$= \frac{2}{1+x}-\frac{1}{1-x}+\frac{3x}{(1-x)(1+x)}$$

$$= \frac{2(1-x)-(1+x)+3x}{(1-x)(1+x)}$$

$$= \frac{1}{(1-x)(1+x)}$$

Example 3

$$\frac{1}{x^2-4}+\frac{1}{x^2+x-6}-\frac{2}{x^2+5x+6}$$

$$= \frac{1}{(x-2)(x+2)}+\frac{1}{(x+3)(x-2)}-\frac{2}{(x+2)(x+3)}$$

$$= \frac{x+3+x+2-2(x-2)}{(x-2)(x+2)(x+3)}$$

$$= \frac{9}{(x-2)(x+2)(x+3)}$$

Example 4

$$\frac{3x}{2-3x+x^2}+\frac{4}{1-x}-\frac{6}{2-x}$$

$$= \frac{3x}{(2-x)(1-x)}+\frac{4}{1-x}-\frac{6}{2-x}$$

$$= \frac{3x+4(2-x)-6(1-x)}{(2-x)(1-x)}$$

$$= \frac{2+5x}{(2-x)(1-x)}$$

Example 5

$$\frac{a+2}{a-2}+\frac{4}{4-a^2}-1$$
$$=\frac{a+2}{a-2}-\frac{4}{a^2-4}-1$$
$$=\frac{a+2}{a-2}-\frac{4}{(a-2)(a+2)}-1$$
$$=\frac{(a+2)^2-4-(a-2)(a+2)}{(a-2)(a+2)}$$
$$=\frac{a^2+4a+4-4-(a^2-4)}{(a-2)(a+2)}$$
$$=\frac{4(a+1)}{(a-2)(a+2)}$$

Example 6

$$\frac{1}{4-\dfrac{3}{2+\dfrac{x}{1-x}}}$$
$$=\frac{1}{4-\dfrac{3}{\dfrac{2(1-x)+x}{1-x}}}$$
$$=\frac{1}{4-\dfrac{3}{\dfrac{2-x}{1-x}}}$$
$$=\frac{1}{4-\dfrac{3(1-x)}{2-x}}$$
$$=\frac{1}{\dfrac{4(2-x)-3(1-x)}{(2-x)}}=\frac{1}{\dfrac{8-4x-3+3x}{2-x}}$$
$$=\frac{1}{\dfrac{5-x}{2-x}}=\frac{2-x}{5-x}$$

Example 7

Simplify

$$\frac{x^2-25}{x^2+3x-10}\times\frac{x^2-4}{x^2-3x-10}\times\frac{x+1}{x^2+3x}$$

$$= \frac{(x-5)(x+5)}{(x+5)(x-2)}\times\frac{(x-2)(x+2)}{(x-5)(x+2)}\times\frac{x+1}{x(x+3)}$$

$$= \frac{x+1}{x(x+3)}$$

Example 8

$$\frac{x^2-3x+2}{(x-3)}\div\frac{x^2-1}{2x^2-6x}$$

$$= \frac{x^2-3x+2}{(x-3)}\times\frac{2x^2-6x}{x^2-1}$$

$$= \frac{(x-1)(x-2)}{(x-3)}\times\frac{2x(x-3)}{(x-1)(x+1)}$$

$$= \frac{2x(x-2)}{x+1}$$

Example 9

Given that $x=\frac{1+y}{2y-1}$ **and** $y=\frac{1+2z}{1-z}$. **Find** z **in terms of** x **only.**

Now in the first equation

$x=\frac{1+y}{2y-1}$, **substituting** $y=\frac{1+2z}{1-z}$, **we get**

$$x=\frac{1+\frac{1+2z}{1-z}}{2\left(\frac{1+2z}{1-z}\right)-1}$$

$$x=\frac{\frac{(1-z)+(1+2z)}{1-z}}{\frac{2(1+2z)-(1-z)}{1-z}}$$

$$x = \frac{\frac{2+z}{1-z}}{\frac{1+5z}{1-z}}$$

$$x = \frac{2+z}{1+5z}$$

$$x(1+5z) = 2+z$$
$$x+5xz = 2+z$$
$$z(5x-1) = 2-x$$
$$z = \frac{2-x}{5x-1}$$

Exercise 3.1

Simplify.

1 $\frac{x}{2x-6}+\frac{3}{6-2x}+\frac{x}{2}$

2 $\frac{6}{x^2+2x-8}+\frac{7}{10-3x-x^2}$

3 $\frac{3}{x^2+2x-15}-\frac{1}{x^2-x-6}-\frac{2}{x^2+7x+10}$

4 $\frac{2x}{x^2-2x-3}+\frac{1}{x^2-1}+\frac{x}{x^2-4x+3}$

5 $x-\frac{1}{1-x}-\frac{x^2-3x-2}{x^2-1}$

6 $\frac{1}{x^2-5x+6}-\frac{2}{x^2-4x+3}+\frac{1}{x^2-3x+2}$

7 $\frac{1}{2x-1}-\frac{2x}{4x^2-1}-\frac{1}{2x^2-3x+1}$

8 $\frac{a-2}{a^2-9a+20}-\frac{a+2}{a^2-a-12}$

9 $\frac{a-2}{a+2}+\frac{a+2}{a-2}-\frac{a^2+4}{a^2-4}$

10 $\frac{1}{x^2-1}-\frac{1}{2x^2-6x+4}+\frac{3}{2x^2-2x-4}$

11. $\dfrac{a^2}{(a-b)(a-c)}+\dfrac{b^2}{(b-c)(b-a)}+\dfrac{bc}{(c-a)(c-b)}$

12. $\dfrac{x-a}{x-b}+\dfrac{x-b}{x-a}-\dfrac{(a-b)^2}{(x-a)(x-b)}$

13. $\dfrac{a^2+3a+2}{a^2-4a-12}\times\dfrac{a^2-7a+6}{a^2-4}$

14. $\dfrac{a^3+b^3}{a(a^2-b^2)}\times\dfrac{a+b}{a-b}\times\dfrac{a^2-ab}{(a+b)^2}$

15. $\dfrac{1}{a^2+ab+b^2}\times\dfrac{2a}{a^3+b^3}\times\dfrac{a^4+a^2b^2+b^4}{4a^2}$

16. $\left(\dfrac{a}{a-1}-\dfrac{a+1}{a}\right)\div\left(\dfrac{a}{a+1}-\dfrac{a-1}{a}\right)$

17. $\left(2-\dfrac{y^2+z^2-x^2}{yz}\right)\div\left(2+\dfrac{x^2+y^2-z^2}{xy}\right)$

18. $\left(\dfrac{a^2+b^2}{a^2-b^2}-\dfrac{a^2-b^2}{a^2+b^2}\right)\div\left(\dfrac{a+b}{a-b}-\dfrac{a-b}{a+b}\right)$

19. **If** $y=x+\dfrac{1}{x}$ **and** $z=y-\dfrac{1}{y}$, **find** z **in terms of** x.

20. **If** $y=\dfrac{1-t}{1+t}$, **express** $\dfrac{1-y^2}{1+y^2}$ **in terms of** t.

21. **If** $x=\dfrac{1+a}{1-a}$ **and** $y=\dfrac{1-a}{1+a}$ **find** $\dfrac{x-y}{1+xy}$ **in terms of** a.

22. **If** $a=\dfrac{2b+1}{b-1}$, $b=\dfrac{c+1}{2c-1}$, **express** $\dfrac{2q+1}{q-1}$ **in terms of** x **only.**